3.

Service models

Iaas

=> Physical data centre

=> Networking & Storage Infrastructure

=> Security & availability

=> manage os, application & data

paas

=> Infrastructure & hardware

=> operating system

=> Runtime environment & middleware

=> eg: Azure

Saas

=> Application Software

=> Data Storage management

=> manage own data

=> usage policies

eg: Salesforce

Xaas

=> Refer to any Cloud Service

=> provide specialized Cloud Service

=> ensure security performance

=> handle update security

EVOLUTION OF CLOUD COMPUTING

Hardware Evolution

=> From mainframe (1960's - 1970's) to personal computer

=> Rise virtualization and multicore-processor (2000's)

=> high performance services SSDS, CWC & scalable

Distributed System

=> From single to machine system

=> peer to peer grid computing

=> Scalable

Service delivery

=> Utility computing

=> Emergence of IaaS, PaaS, SaaS

=> Serverless & edge computing

Internet software evolution

=> Static web page

=> Dynamic Web

=> Cloud Native

Service models

IaaS

4)

Cloud Computing & Virtualization

Virtualization

=> enable multitenancy
=> provide on demand
=> Reduce hardware cost
=> Isolation & security

Web Service

=> Connects application & service across the cloud
=> Supports hybrid and multicloud environment
Eg: HTTP, REST

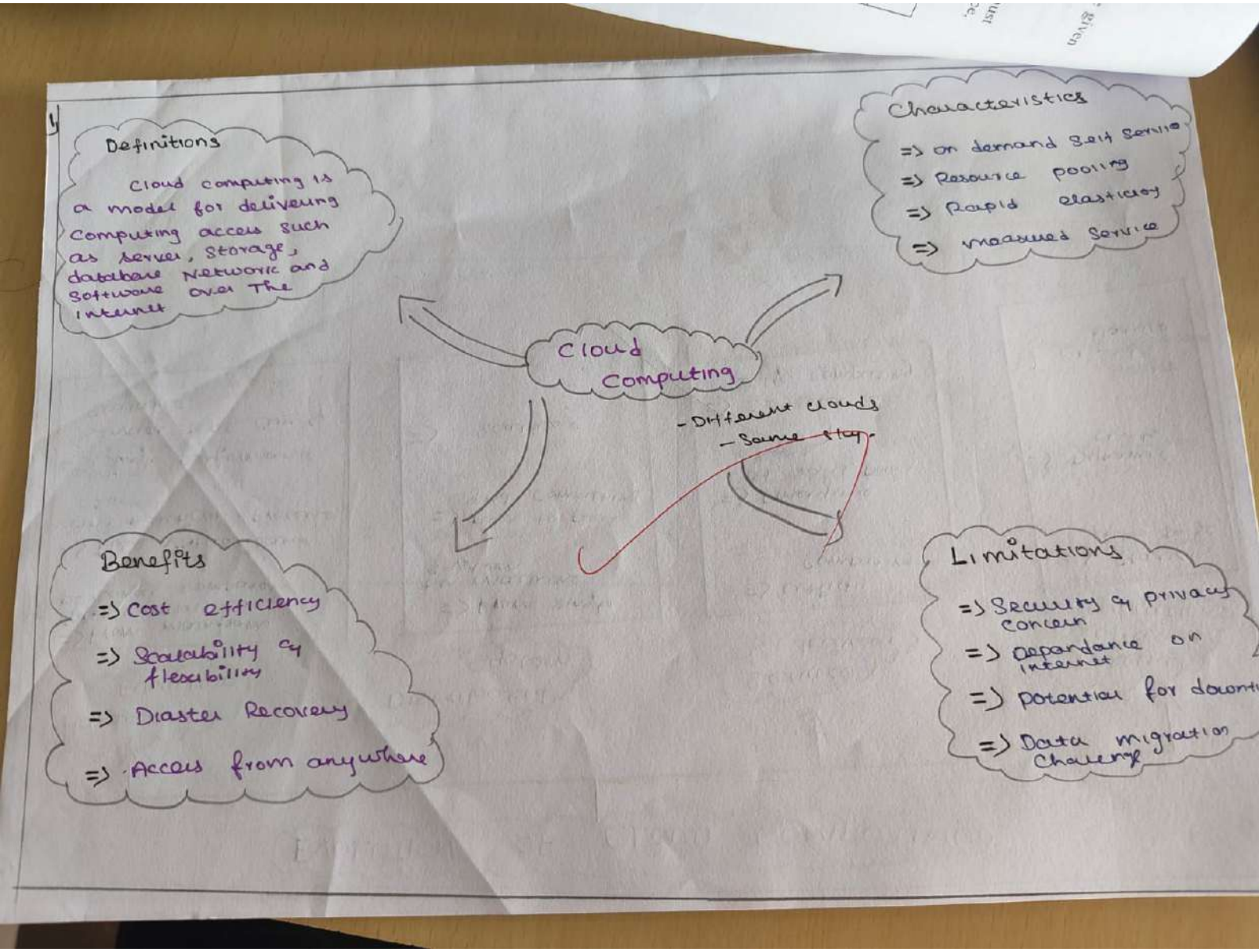
Enable cloud Deployment

=> Virtualization abstract hardware, allowing clouds provider to run multiple instance
=> Together they simplify setup, reduce complexity & speedup

Enable Scalability

=> Virtualization allow dynamic provisioning and live migration of virtual machine
=> web service enable seamless scaling through API load balance

13/10/25


Definitions
Cloud computing is a model for delivering computing access such as server, storage, database Network and software over The internet
Characteristics
=> on demand Self Servic
=> Resource pooling
=> Rapid elasticity
=> measured service
Cloud Computing
- Different clouds
- Same step
Benefits
=> Cost efficiency
=> Scalability & flexibility
=> Diaster Recovery
=> Access from anywhere
Limitations
=> Security & privacy concern
=> Depandance on internet
=> potential for downt
=> Data migration Challeng